

Foodie Saver: Finding the Best Bite for your Buck

*An Innovative solution for the food
enthusiasts*

END-TO-END-DATA-SCIENCE-PROJECT TASK-3

COMPANY: CODTECH IT SOLUTIONS

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DOMAIN: Data Science

DURATION: 4 Weeks

MENTOR: NEELA SANTOSH

Foodie Saver is a comprehensive food discovery and ordering platform that goes beyond the ordinary. It seamlessly integrates three key elements — price comparison for food ordering, mood-based meal suggestions, and ingredient-based recipe creation. Whether you're craving the ease of ordering, seeking culinary inspiration, or planning a home-cooked meal, Foodie Saver is an all-in-one solution.

Objective:

Enhanced Food Discovery: Simplify the process of discovering new and exciting dishes tailored to your mood and preferences. **Effortless Ordering:** Compare prices across popular food ordering platforms to ensure you get the best value for your money.

Creative Cooking: Make cooking at home an enjoyable experience by receiving personalized recipes based on the ingredients you have on hand.

Goals:

User Satisfaction: Prioritize user satisfaction by offering a userfriendly interface, personalized recommendations, and reliable price information.

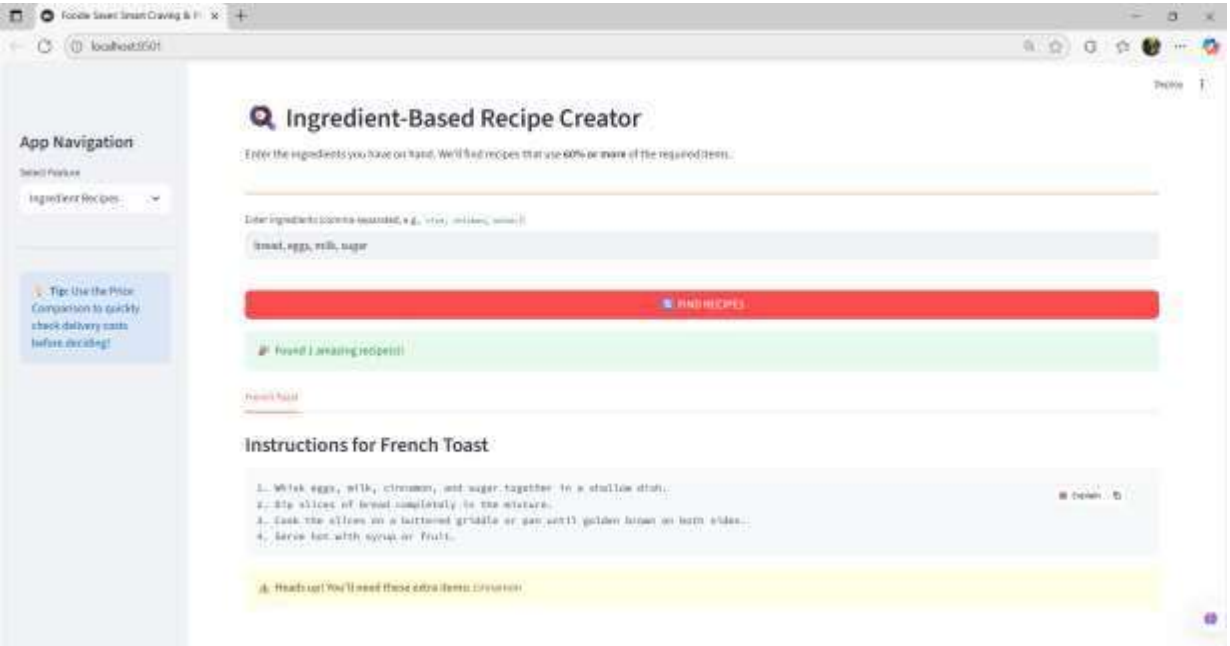
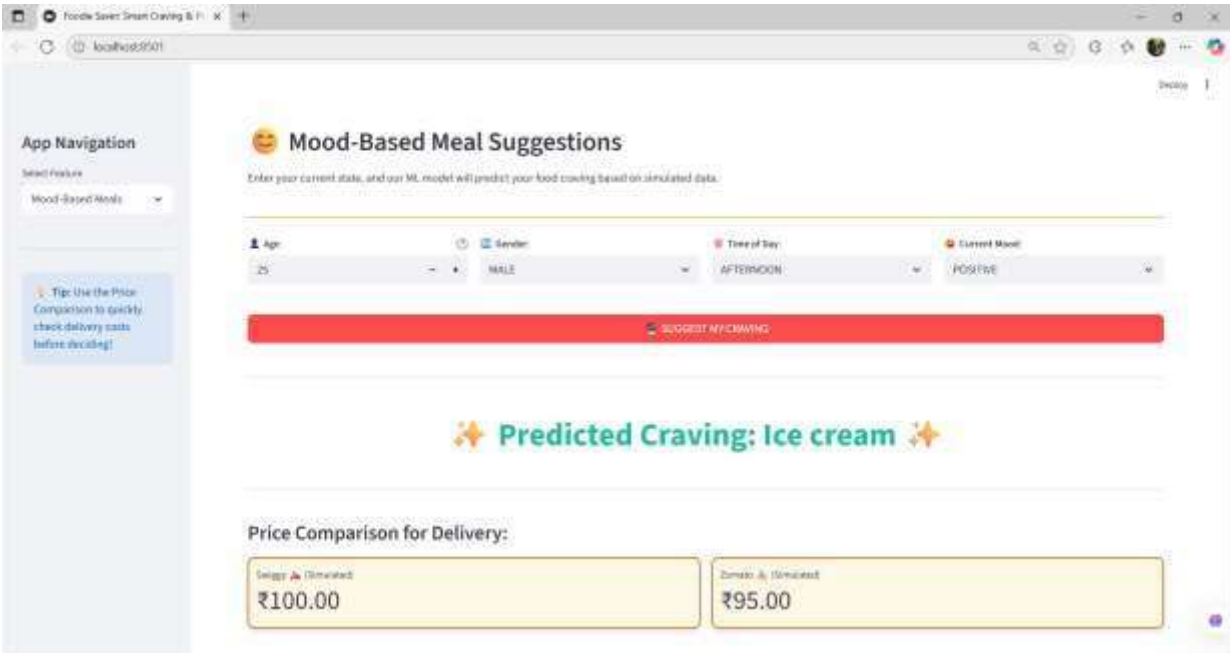
Market Impact: Establish Foodie Saver as a prominent player in the food tech industry, filling the gaps left by existing solutions. **Culinary Exploration:** Inspire users to explore new flavours, cuisines, and cooking techniques, fostering a sense of culinary adventure.

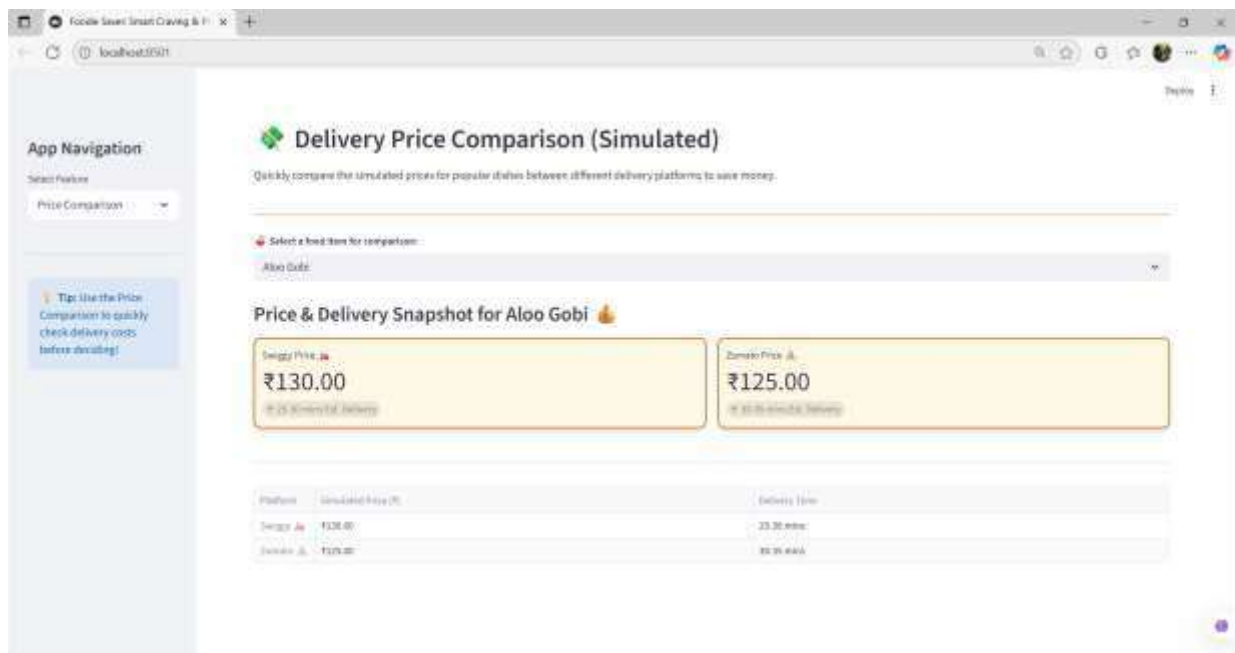
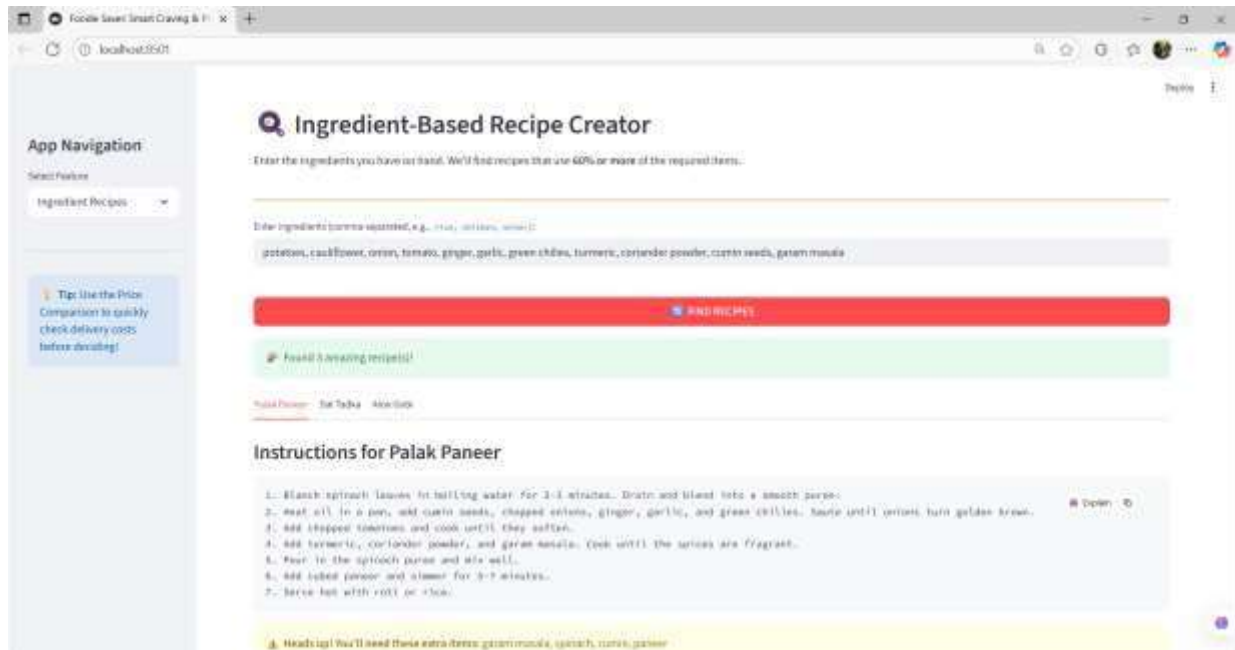
Partnerships: Form strategic partnerships with popular food ordering platforms to provide users with an extensive and diverse range of choices.

Monetization: Implement sustainable and ethical monetization strategies, ensuring the app's longevity and continuous improvement.

Prototype:
FRONT-END:

MANUAL integration INPUT:





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Food Allergies

- ☐ Eggs
- ☐ Dairy
- ☐ Soybeans
- ☐ Peanuts
- ☐ Fish
- ☐ Meat
- ☐ Wheat
- ☐ None

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Medical Conditions

- ☐ Diabetes
- ☐ Pre-diabetes
- ☐ Cholesterol
- ☐ Hypertension
- ☐ Stress
- ☐ Sleep Issues
- ☐ Thyroid
- ☐ None

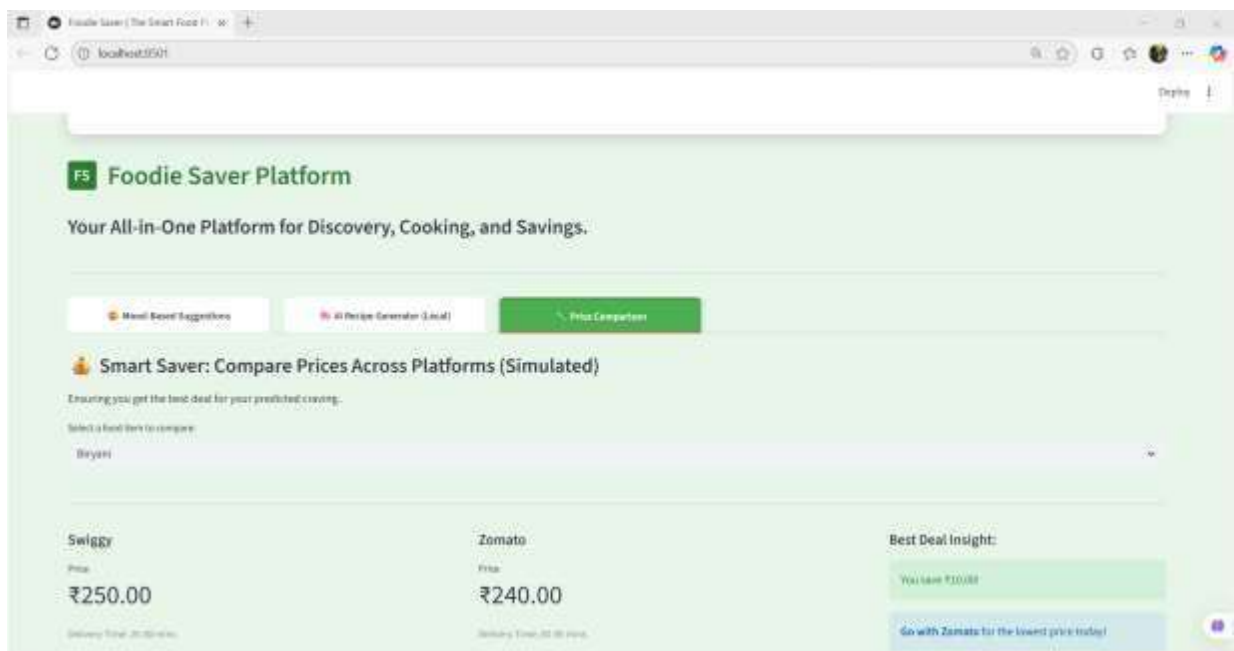
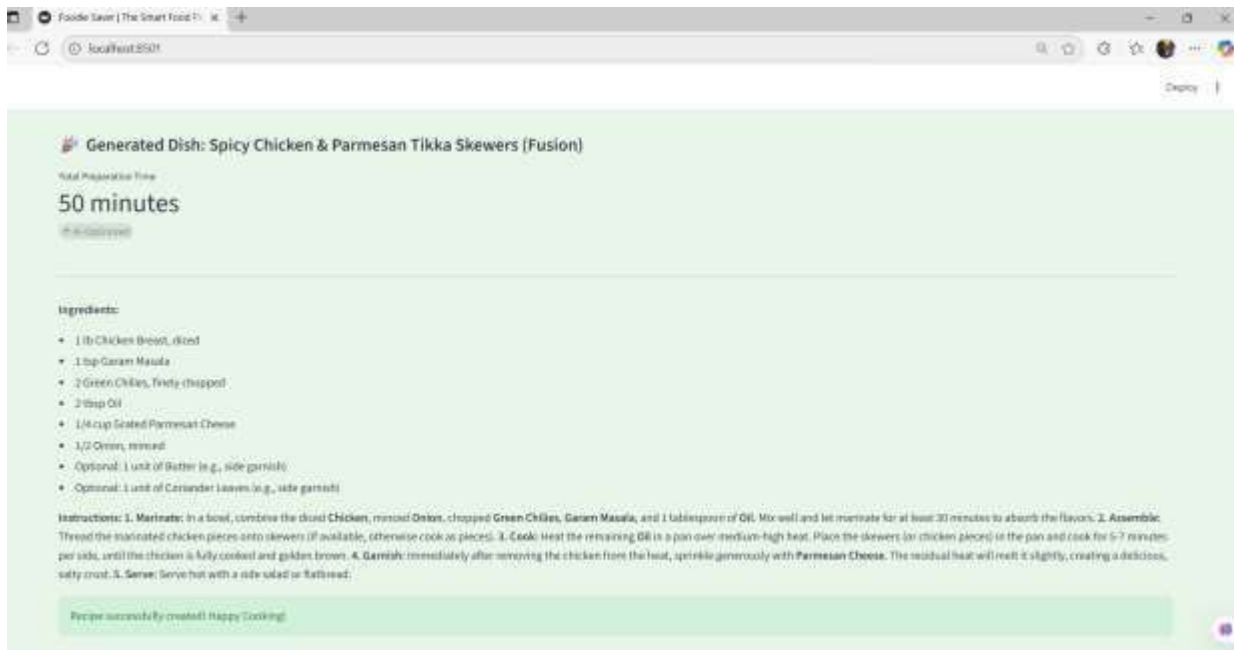
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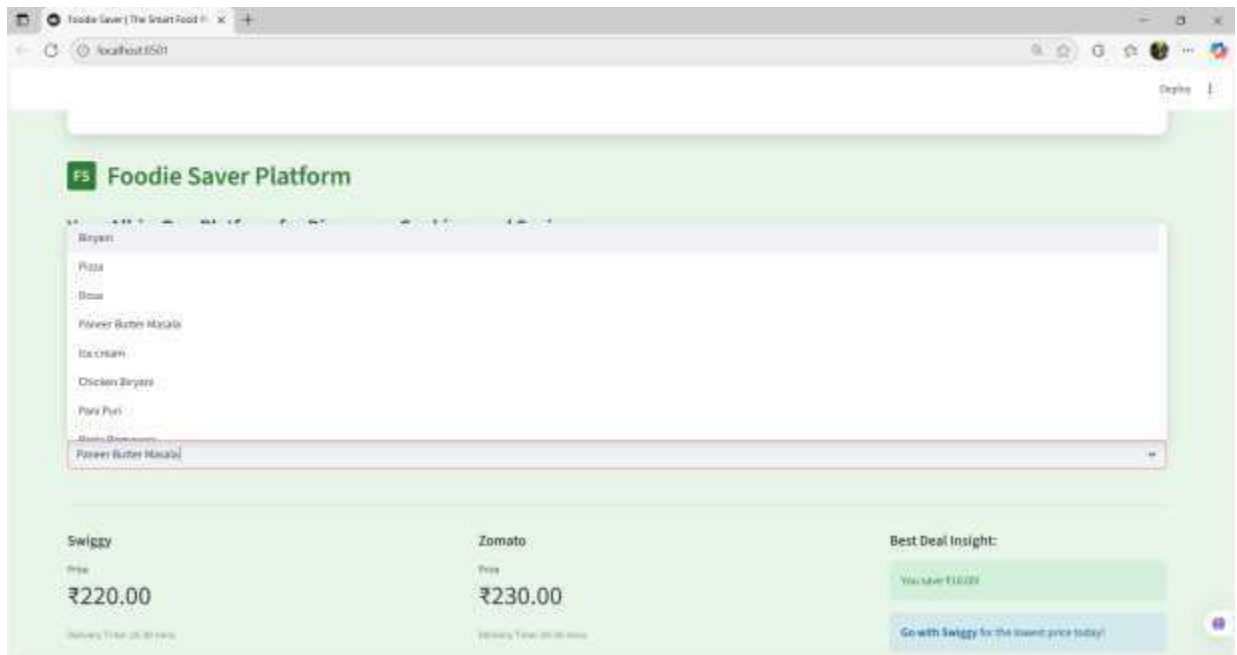
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AI DRIVEN AUTO RECIPES OBTAINED:

The screenshot shows a web browser window with the URL `localhost:5001`. The page has a light green background. At the top, there's a search bar. Below it, the text "Foodie Saver Platform" is displayed, followed by the tagline "Your All-in-One Platform for Discovery, Cooking, and Savings." There are three buttons: "Mood-Based Suggestions" (green), "AI Recipe Generator (Local)" (pink), and "Price Comparison" (light green). The "Mood Analyzer" section is active, with the text "Tell us how you're feeling, let the AI suggest your next craving!". It includes a slider for "Enter your age" (set to 25), a dropdown for "Time of the day" (set to MORNING), a dropdown for "Your gender" (set to MALE), and a dropdown for "Your mood" (set to POSITIVE). A large red button labeled "Suggest My Craving" is at the bottom.

The screenshot shows the same web browser window, but now displaying the results of the mood analysis. The "Mood Analyzer" section is still visible, but the "Your mood" dropdown is now set to "NEUTRAL". Below the "Suggest My Craving" button, there's a section titled "Your AI-Predicted Craving:" with a green checkmark. It shows "Predicted Mood: Coffee" and a note "Tailored to your mood!". Below this is a "Simulated Price Comparison:" section with two boxes. The left box is blue and labeled "Twigg", showing "Price: N/A" and "Time: 25 mins". The right box is green and labeled "Zomato", showing "Price: N/A" and "Time: 30 mins".





BACK-END development:

```
def suggest_recipe():
    available_ingredients = []
    print('Please enter the available ingredients one by one. Type \'done\' when finished.')
    while True:
        ingredient = input('Enter an ingredient: ').strip().lower()
        if ingredient == 'done':
            break
        available_ingredients.append(ingredient)
    recipes = {
        "Pasta Primavera": {
            "ingredients": ["pasta", "assorted vegetables", "olive oil", "garlic", "parmesan cheese"],
            "instructions": "1. Cook pasta according to package instructions. \n2. In a pan, sauté assorted vegetables and minced garlic in olive oil until tender. \n3. Toss cooked pasta with the vegetable mixture. \n4. Serve hot, garnished with grated parmesan cheese."
        },
        "Vegetable Stir Fry": {
            "ingredients": ["assorted vegetables", "soy sauce", "sesame oil", "garlic", "ginger", "rice"],
            "instructions": "1. Cook rice according to package instructions. \n2. In a wok or skillet, stir-fry assorted vegetables, minced garlic, and ginger in sesame oil until tender. \n3. Add cooked rice and soy sauce, stir-fry until well combined. \n4. Serve hot."
        },
        "Caprese Salad": {
            "ingredients": ["tomatoes", "fresh mozzarella", "fresh basil leaves", "balsamic vinegar", "olive oil"],
            "instructions": "1. Slice tomatoes and fresh mozzarella cheese into rounds. \n2. Arrange alternating slices of tomatoes, mozzarella, and basil leaves on a serving plate. \n3. Drizzle with balsamic vinegar and olive oil. \n4. Sprinkle with salt and pepper to taste. \n5. Serve immediately."
        },
        "Tomato Basil Pasta": {
            "ingredients": ["pasta", "tomatoes", "garlic", "fresh basil leaves", "olive oil"],
            "instructions": "1. Cook pasta according to package instructions. \n2. In a pan, sauté chopped tomatoes and minced garlic in olive oil until tomatoes break down. \n3. Toss cooked pasta with the tomato mixture and torn fresh basil leaves. \n4. Serve hot."
        },
        "Egg Fried Rice": {
            "ingredients": ["rice", "eggs", "assorted vegetables", "soy sauce", "sesame oil", "garlic"],
            "instructions": "1. Cook rice according to package instructions and let it cool. \n2. In a wok or skillet, scramble eggs in sesame oil. \n3. Add minced garlic and stir-fry assorted vegetables until tender. \n4. Add cooked rice and soy sauce, stir-fry until heated through. \n5. Serve hot."
        },
        "Chana Masala": {
            "ingredients": ["chickpeas", "onion", "tomatoes", "ginger", "garlic", "coriander", "cumin", "garam masala", "turmeric"],
        }
    }
```

```

"Instructions": "1. Heat oil in a pan and add chopped onions. Sauté until golden brown. \n2. Add ginger and garlic paste, sauté until the raw smell disappears. \n3. Add chopped tomatoes and cook until they turn mushy. \n4. Add spices (coriander, cumin, garam masala, turmeric) and sauté for a minute. \n5. Add boiled chickpeas and salt. Mix well. \n6. Cook for 10-15 minutes on medium heat. \n7. Garnish with chopped coriander leaves and serve hot."
},
"Paneer Tikka Masala": {
  "Ingredients": ["paneer", "onion", "bell peppers", "yogurt", "tomato puree", "ginger-garlic paste", "kasuri methi", "garam masala", "turmeric", "coriander powder", "cumin powder"],
  "Instructions": "1. Marinate paneer cubes in yogurt, ginger-garlic paste, turmeric, and salt. Let it sit for 30 minutes. \n2. Grill or roast marinated paneer cubes until golden brown. \n3. In a pan, heat oil and add chopped onions. Cook until they turn translucent. \n4. Add ginger-garlic paste and sauté until the raw smell disappears. \n5. Add tomato puree, coriander powder, cumin powder, garam masala, and salt. Cook until the oil separates. \n6. Add grilled paneer cubes, bell peppers, and kasuri methi. Mix well. \n7. Cook for another 5-7 minutes. \n8. Garnish with cream and coriander leaves. Serve hot with naan or rice."
},
"Chicken Curry": {
  "Ingredients": ["chicken", "onion", "tomatoes", "ginger", "garlic", "chili powder", "turmeric", "coriander powder", "cumin powder", "garam masala", "coconut milk"],
  "Instructions": "1. Heat oil in a pan and add chopped onions. Sauté until golden brown. \n2. Add ginger and garlic paste, sauté until the raw smell disappears. \n3. Add chopped tomatoes and cook until they turn mushy. \n4. Add chili powder, turmeric, coriander powder, cumin powder, garam masala, and salt. Cook until the oil separates. \n5. Add chicken pieces and coat well with the masala. \n6. Add water and cook until the chicken is tender. \n7. Add coconut milk and simmer for another 5-7 minutes. \n8. Garnish with coriander leaves. Serve hot with rice or roti."
},
"Palak Paneer": {
  "Ingredients": ["paneer", "spinach", "onion", "tomato", "ginger", "garlic", "green chilies", "garam masala", "cumin", "coriander powder", "turmeric"],
  "Instructions": "1. Blanch spinach leaves in boiling water for 2-3 minutes. Drain and blend into a smooth puree. \n2. Heat oil in a pan, add cumin seeds, chopped onions, ginger, garlic, and green chilies. Sauté until onions turn golden brown. \n3. Add chopped tomatoes and cook until they soften. \n4. Add turmeric, coriander powder, and garam masala. Cook until the spices are fragrant. \n5. Pour in the spinach puree and mix well. Cook for a few minutes. \n6. Add cubed paneer and simmer for 5-7 minutes. \n7. Serve hot with roti or rice."
},
"Dal Tadka": {
  "Ingredients": ["yellow lentils", "onion", "tomato", "ginger", "garlic", "green chilies", "turmeric", "cumin seeds", "mustard seeds", "asafoetida", "garam masala"],
  "Instructions": "1. Rinse lentils and cook with water, turmeric, salt, and chopped tomatoes until soft. Mash slightly. \n2. Heat ghee in a pan, add cumin seeds, mustard seeds, asafoetida, chopped onions, ginger, garlic, and green chilies. Sauté until onions turn golden brown. \n3. Add cooked lentils and adjust consistency by adding water if needed. \n4. Add garam masala and simmer for a few minutes. \n5. Garnish with coriander leaves and serve hot with rice or roti."
},
"Aloo Gobi": {
  "Ingredients": ["potatoes", "cauliflower", "onion", "tomato", "ginger", "garlic", "green chilies", "turmeric",

```

UNRECOGNIZED

```
    "coriander powder", "cumin seeds", "garam masala"],
    "instructions": "1. Heat oil in a pan, add cumin seeds, chopped onions, ginger, garlic, and green chilies. Sauté until onions turn golden brown. \n2. Add chopped tomatoes and cook until they soften. \n3. Add turmeric, coriander powder, and garam masala. Cook until the spices are fragrant. \n4. Add diced potatoes and cauliflower florets. Mix well. \n5. Cover and cook until vegetables are tender. \n6. Garnish with coriander leaves and serve hot with roti or rice."
},
"Vegetable Biryani": {
    "ingredients": ["rice", "assorted vegetables", "onion", "tomato", "ginger", "garlic", "green chilies", "yogurt", "mint leaves", "coriander leaves", "garam masala", "turmeric", "saffron"],
    "instructions": "1. Soak rice in water for 30 minutes. Drain and set aside. \n2. Heat oil in a pan, add cumin seeds, chopped onions, ginger, garlic, and green chilies. Sauté until onions turn golden brown. \n3. Add chopped tomatoes and cook until they soften. \n4. Add assorted vegetables, yogurt, mint leaves, coriander leaves, garam masala, turmeric, and saffron. Mix well. \n5. In a separate pot, layer soaked rice and vegetable mixture. \n6. Cover and cook on low heat until rice is cooked and aromatic. \n7. Serve hot with raita."
},
"Chicken Biryani": {
    "ingredients": ["rice", "chicken", "onion", "tomato", "ginger", "garlic", "green chilies", "yogurt", "mint leaves", "coriander leaves", "garam masala", "turmeric", "saffron"],
    "instructions": "1. Soak rice in water for 30 minutes. Drain and set aside. \n2. Marinate chicken pieces in yogurt, ginger-garlic paste, green chilies, turmeric, and garam masala. Let it sit for 30 minutes. \n3. Heat oil in a pan, add chopped onions and cook until golden brown. \n4. Add chopped tomatoes and cook until they soften. \n5. Add marinated chicken and cook until it is partially done. \n6. In a separate pot, layer soaked rice and chicken mixture. Add mint leaves, coriander leaves, and saffron. \n7. Cover and cook on low heat until rice is cooked and chicken is tender. \n8. Serve hot with raita."
},
"Pani Puri": {
    "ingredients": ["semolina flour", "all-purpose flour", "potatoes", "chickpeas", "tamarind pulp", "mint leaves", "green chilies", "chaat masala", "cumin powder", "coriander powder", "black salt"],
    "instructions": "1. Prepare a dough using semolina flour, all-purpose flour, and water. Let it rest for 30 minutes. \n2. Roll out small puris from the dough and deep fry until they puff up. \n3. Boil and mash potatoes, and prepare a spicy water by mixing tamarind pulp, mint leaves, green chilies, chaat masala, cumin powder, coriander powder, and black salt. \n4. Make a hole in each puri and stuff with mashed potatoes and chickpeas. \n5. Dip the stuffed puris in the spicy water and serve immediately."
},
"Masoor Dal": {
    "ingredients": ["red lentils", "onion", "tomato", "ginger", "garlic", "green chilies", "turmeric", "cumin seeds", "mustard seeds", "curry leaves", "asafoetida", "garam masala"],
    "instructions": "1. Rinse lentils and cook with water, turmeric, salt, and chopped tomatoes until soft. Mash slightly. \n2. Heat oil in a pan, add cumin seeds, mustard seeds, asafoetida, chopped onions, ginger, garlic, green chilies, and curry leaves. Sauté until onions turn golden brown. \n3. Add cooked lentils and adjust consistency by adding water if needed. \n4. Add garam masala and simmer for a few minutes. \n5. Garnish with coriander leaves and serve hot with rice or roti."
},
"Vegetable Pakora": {
```

```

        "ingredients": ["gram flour", "assorted vegetables", "onion", "ginger", "garlic", "green chilies", "coriander leaves",
                        "cumin seeds", "carom seeds", "turmeric", "chili powder", "garam masala"],
        "instructions": "1. Mix gram flour, chopped onions, ginger, garlic, green chilies, coriander leaves, cumin seeds, carom seeds, turmeric, chili powder, garam masala, and salt in a bowl. Add water to make a thick batter. \n2. Heat oil in a deep frying pan. Drop spoonfuls of the batter into the hot oil and fry until golden brown and crispy. \n3. Serve hot with mint chutney or tamarind chutney."
    }
}

threshold = 0.6 # 60%

suggested_recipes = []
for recipe, details in recipes.items():
    required_ingredients = details["ingredients"]
    matching_ingredients = sum(ingredient in available_ingredients for ingredient in required_ingredients)
    if matching_ingredients / len(required_ingredients) >= threshold:
        suggested_recipes.append((recipe, details["instructions"]))

if suggested_recipes:
    print("\nBased on your available ingredients, you can try making:")
    for recipe, instructions in suggested_recipes:
        print("\nRecipe: " + recipe)
        print("Instructions:")
        print(instructions)
        print("="*50)
else:
    print("\nSorry, with the available ingredients, I couldn't find a suitable recipe.")

suggest_recipe()

```


Untitled1

Please enter the available ingredients one by one. Type 'done' when finished.

Enter an ingredient: chicken

Enter an ingredient: rice

Enter an ingredient: ginger

Enter an ingredient: garlic

Enter an ingredient: onion

Enter an ingredient: tomato

Enter an ingredient: yogurt

Enter an ingredient: mint leaves

Enter an ingredient: turmeric

Enter an ingredient: saffron

Enter an ingredient: done

Based on your available ingredients, you can try making:

Recipe: Vegetable Biryani

Instructions:

1. Soak rice in water for 30 minutes. Drain and set aside.
2. Heat oil in a pan, add cumin seeds, chopped onions, ginger, garlic, and green chillies. Sauté until onions turn golden brown.
3. Add chopped tomatoes and cook until they soften.
4. Add assorted vegetables, yogurt, mint leaves, coriander leaves, garam masala, turmeric, and saffron. Mix well.
5. In a separate pot, layer soaked rice and vegetable mixture.
6. Cover and cook on low heat until rice is cooked and aromatic.
7. Serve hot with raita.

Recipe: Chicken Biryani

Instructions:

1. Soak rice in water for 30 minutes. Drain and set aside.
2. Marinate chicken pieces in yogurt, ginger-garlic paste, green chillies, turmeric, and garam masala. Let it sit for 30 minutes.
3. Heat oil in a pan, add chopped onions and cook until golden brown.
4. Add chopped tomatoes and cook until they soften.
5. Add marinated chicken and cook until it is partially done.
6. In a separate pot, layer soaked rice and chicken mixture. Add mint leaves, coriander leaves, and saffron.
7. Cover and cook on low heat until rice is cooked and chicken is tender.
8. Serve hot with raita.

```
import pandas as pd
```

```
data = pd.read_csv("Untitled form.csv")
```

```
data.head()
```

	Timestamp	NAME	AGE	GENDER	TIME OF THE DAY	MOOD	CRAVINGS	ALLERGIES IF ANY:
0	2025/04/29 7:22:14 pm GMT+5:30	Aishwarya	20-30	FEMALE	MORNING	NEUTRAL	Pancakes	Soya
1	2025/04/29 7:22:42 pm GMT+5:30	Ritesh	10-20	MALE	AFTERNOON	POSITIVE	Biryani	Nil
2	2025/04/29 7:23:22 pm GMT+5:30	Sathish	20-30	MALE	EVENING	POSITIVE	Pizza	Nuts
3	2025/04/29 7:24:33 pm GMT+5:30	Shamav	30-40	MALE	EVENING	NEGATIVE	Ice cream	Prawns
4	2025/04/29 7:25:02 pm GMT+5:30	Mona shree	10-20	FEMALE	AFTERNOON	POSITIVE	Biryani	Peanuts

```
[7]: data = data.drop(columns=["Timestamp", "NAME"])
```

```
[11]: data.head()
```

	AGE	GENDER	TIME OF THE DAY	MOOD	CRAVINGS	ALLERGIES IF ANY:
0	20-30	FEMALE	MORNING	NEUTRAL	Pancakes	Soya
1	10-20	MALE	AFTERNOON	POSITIVE	Biryani	Nil
2	20-30	MALE	EVENING	POSITIVE	Pizza	Nuts
3	30-40	MALE	EVENING	NEGATIVE	Ice cream	Prawns
4	10-20	FEMALE	AFTERNOON	POSITIVE	Biryani	Peanuts

```
[9]: # Map age ranges to numerical values
age_mapping = {
    '<10': 0,
    '10-20': 1,
    '20-30': 2,
    '30-40': 3,
    '40-50': 4,
    '50-60': 5,
    '>60': 6
}
data['AGE'] = data['AGE'].map(age_mapping)

[10]: data.shape

[10]: (40, 6)

[11]: data.columns

[11]: Index(['AGE', 'GENDER', 'TIME OF THE DAY', 'MOOD', 'CRAVINGS',
        'ALLERGIES IF ANY:'],
        dtype='object')

[12]: from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import classification_report, accuracy_score

[13]: # Encode categorical features
label_encoder = LabelEncoder()
data['GENDER'] = label_encoder.fit_transform(data['GENDER'])
data['TIME OF THE DAY'] = label_encoder.fit_transform(data['TIME OF THE DAY'])
data['MOOD'] = label_encoder.fit_transform(data['MOOD'])
data['ALLERGIES IF ANY:'] = label_encoder.fit_transform(data['ALLERGIES IF ANY:'])
# Handle missing values if any
data.fillna(0, inplace=True) # Using 0 as a simple fill strategy

[14]: # 2. Model Training
X = data[['AGE', 'GENDER', 'TIME OF THE DAY', 'MOOD']]
y = data['CRAVINGS']

# Split the data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

[15]: # Scale the features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

[16]: # Initialize and train the KNN model
k = 1
knn_model = KNeighborsClassifier(n_neighbors=k)
knn_model.fit(X_train_scaled, y_train)

[16]: • KNeighborsClassifier
KNeighborsClassifier(n_neighbors=1)

[17]: y_pred = knn_model.predict(X_test_scaled)
```

```
[10]: # 3. User Input and Prediction
print("... Craving Prediction ...")
try:
    # The script will wait here for your first input
    user_age = int(input("Enter age: "))

    if user_age < 10: user_age_range = '<10'
    elif user_age < 20: user_age_range = '10-20'
    elif user_age < 30: user_age_range = '20-30'
    elif user_age < 40: user_age_range = '30-40'
    elif user_age < 50: user_age_range = '40-50'
    elif user_age < 60: user_age_range = '50-60'
    else: user_age_range = '>60'

    user_age_encoded = age_mapping[user_age_range]

    # Then it will wait here for your second input
    user_gender = input("Enter gender (MALE/FEMALE/OTHERS): ").upper()
    # And finally, it will wait here for your third input
    user_time_of_day = input("Enter time of the day (MORNING/AFTERNOON/EVENING/NIGHT): ").upper()
    user_mood = input("Enter mood (POSITIVE/NEGATIVE/NEUTRAL): ").upper()

    # Mappings for user inputs
    gender_mapping = {'MALE': 0, 'FEMALE': 1, 'OTHERS': 2}
    time_of_day_mapping = {'MORNING': 0, 'AFTERNOON': 1, 'EVENING': 2, 'NIGHT': 3}
    mood_mapping = {'POSITIVE': 0, 'NEGATIVE': 1, 'NEUTRAL': 2}

    user_gender_encoded = gender_mapping.get(user_gender, -1)
    user_time_of_day_encoded = time_of_day_mapping.get(user_time_of_day, -1)
    user_mood_encoded = mood_mapping.get(user_mood, -1)

    if -1 in [user_gender_encoded, user_time_of_day_encoded, user_mood_encoded]:
        print("Invalid Input. Please check your spelling and try again.")
    else:
        user_input = [[user_age_encoded, user_gender_encoded, user_time_of_day_encoded, user_mood_encoded]]
        user_input_scaled = scaler.transform(user_input)
        predicted_craving = knn_model.predict(user_input_scaled)
        print(f"Predicted craving: {predicted_craving[0]}")
```

```
if -1 in [user_gender_encoded, user_time_of_day_encoded, user_mood_encoded]:
    print("Invalid Input. Please check your spelling and try again.")
else:
    user_input = [[user_age_encoded, user_gender_encoded, user_time_of_day_encoded, user_mood_encoded]]
    user_input_scaled = scaler.transform(user_input)
    predicted_craving = knn_model.predict(user_input_scaled)
    print(f"Predicted craving: {predicted_craving[0]}")
```

```
except ValueError:
    print("Invalid age entered. Please enter a whole number.")
except Exception as e:
    print(f"An unexpected error occurred: {e}")

... Craving Prediction ...
Enter age: 28
Enter gender (MALE/FEMALE/OTHERS): MALE
Enter time of the day (MORNING/AFTERNOON/EVENING/NIGHT): AFTERNOON
Enter mood (POSITIVE/NEGATIVE/NEUTRAL): NEUTRAL
Predicted craving: Coffee
```

```
[11]: import joblib
# knn_model = KNeighborsClassifier(n_neighbors=k)
# knn_model.fit(X_train_scaled, y_train)
# ADD THESE TWO LINES TO SAVE YOUR FILES:
print("Saving model...")
joblib.dump(knn_model, 'knn_model.pkl')
joblib.dump(scaler, 'scaler.pkl')
print("Model and scaler saved!")

Saving model...
Model and scaler saved!
```

```
[11]:
```